

EVALUATING SELF-SUPERVISED PRE-TRAINED VISION TRANSFORMER FOR BRAIN TUMOR CLASSIFICATION WITH HISTOGRAM EQUALIZATION IMAGE ENHANCEMENT

AEDENTRISA YASMANDA PAULINDINO^{1,*}, GREGORIUS NATANAEL ELWIREHARDJA^{2,3}
AND BENS PARDAMEAN^{1,2}

¹Computer Science Department, BINUS Graduate Program – Master of Computer Science

²Bioinformatics and Data Science Research Center

³Computer Science Department, School of Computer Science
Bina Nusantara University

Jl. K. H. Syahdan No. 9, Kemanggisian, Palmerah, Jakarta 11480, Indonesia
{ gregorius.e; bpardamean }@binus.edu

*Corresponding author: aedentrisa.paulindino@binus.ac.id

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ABSTRACT. *Diagnosing brain tumors, benign or malignant, is crucial due to their threat to the vital human brain. This paper presents a novel approach using Self-Supervised Learning (SSL) with the DINOv2 ViT-S14 model to improve brain tumor diagnosis accuracy. This model leverages pre-trained Vision Transformers, including DINOv2, SimCLRv2-ResNet50, and supervised models like ResNet50, VGG-19, and EfficientNet-B0, to improve diagnostic accuracy. Furthermore, the study explores the implementation of image enhancement techniques, such as histogram equalization, to elevate the quality of MRI scans. Results indicate that the SSL model, particularly DINOv2 ViT-S14, outperforms other models with a remarkable precision of 99.79%, recall of 99.89%, and an F1-score of 99.84%. This model excels in addressing data imbalances, as evidenced by the ROC AUC curve. The DINOv2 ViT-S14 model excels in brain tumor classification, achieving high accuracy despite data imbalances and limited data availability. Nonetheless, data imbalance remains a challenge, calling for further exploration into multiclass classification and data augmentation. This innovative approach not only holds promise for broader applications in medical image classification but also highlights the potential of SSL in addressing data imbalances and improving medical diagnostics.*

Keywords: Brain tumor, MRI, Vision Transformer, Transfer learning, Histogram equalization

1. Introduction. The intricate human brain controls cognition, emotions, and bodily functions, housed within the protective skull and cerebrospinal fluid [1]. Brain tumors were the top cause of cancer-related death in children aged 0-14 in the US in 2016, surpassing Leukemia, and ranked third among teenagers and adolescents aged 15-39 [2]. Brain tumors, whether benign or malignant, result from uncontrolled cell growth in the brain or spinal cord, presenting diagnostic and treatment challenges [3]. High mortality rates emphasize the urgency of accurate diagnosis and treatment.

Currently, Magnetic Resonance Imaging (MRI) stands as the gold standard for diagnosing brain tumors, offering detailed insights into tumor characteristics [4]. Computer-Aided Diagnosis (CAD) using traditional machine learning aids radiologists in interpreting MRI scans for early tumor detection [5].

Medical data often has limited availability due to labor-intensive data collection. Self-Supervised Learning (SSL), known for its robustness in addressing data imbalances, is frequently employed in Transfer Learning (TL) [6]. SSL, as proposed [7], enhances feature representation in medical data, improving performance in tasks like diagnosis and

medical image processing [8,9]. SSL leverages unlabeled data to address data limitations and class imbalance, benefiting various medical tasks [10,11]. Among SSL methods, DINO (Knowledge distillation without labels) stands out by consistently outperforming alternative approaches [12].

Numerous studies enhance brain tumor classification using deep learning models, particularly CNNs like VGG-16, VGG-19, DenseNet121, DenseNet201, GoogleNet, ResNet50, and Inception-v3, employing pre-trained models and TL strategies for improved generalization with limited labeled data [13-16]. Notably, TL with VGG-16 is a favored approach, with Singh et al. [17] achieving 90% accuracy. Öztürk and Katar [18] used EfficientNetB0, attaining 99.33% accuracy.

In model comparisons, VGG-19 stood out as a top performer. Sharma et al. [19] reported accuracy and sensitivity rates of 98% and 94.73% when comparing VGG-16, VGG-19, DenseNet121, and DenseNet201. On the other hand, Mehrotra et al. [20] found that TL using AlexNet achieved the highest results at 99.04% accuracy. In a comparative study by Srinivas et al. [21] involving CNN pre-trained models VGG-16, ResNet50, and Inception-v3, VGG-16 outperformed with a 0.96% accuracy rate. Additionally, Yadav et al. [22] employed VGG-16, VGG-19, and ResNet50 CNN models, VGG-19 excelled with 99% accuracy.

To optimize results, researchers modified pre-trained models. Anagun [23] developed an EfficientNetv2-based CNN for brain tumor diagnosis, achieving 99.89% test accuracy and 98.21% AUC. Rastogi et al. [24] combined VGG-19 with additional layers for data augmentation, yielding strong F1-score, recall, accuracy, precision, and ROC AUC scores. Another study by Ratnaparkhi et al. [25] demonstrated that a sequential CNN can reach 99% accuracy.

In summary, CNN techniques for brain tumor classification use pre-trained models and TL to overcome data limitations, boost generalization, and achieve accuracy [13]. Notably, VGG-19, EfficientNetB0, VGG-16, and AlexNet variants consistently perform well in various studies. However, challenges like dataset size and model performance require further research and improvement. This study addresses the challenge by utilizing DINO, a method that consistently outperforms other SSL and supervised approaches [12], presenting a novel brain tumor diagnosis approach with SSL using the DINOv2 ViT model. Brain MRI classification utilizes a pre-trained Vision Transformer (ViT), specifically DINOv2, incorporating a self-distillation approach. Additionally, supervised ViT, SimCLRv2-ResNet50, and supervised ResNet50 are included. These models were compared to baseline models, namely ResNet50, known for its skip connections addressing the vanishing gradient problem [21]; EfficientNetB0, recognized for its efficient use of model parameters [18]; and VGG-19, widely used for its simplicity and effectiveness [26]. The baseline models were duplicated and trained on the same dataset for a comprehensive evaluation. Moreover, the inclusion of image enhancement techniques as a preprocessing step to improve image features [27], specifically histogram equalization [28], is being explored to improve the quality of MRI scans used in diagnosis by improving contrast in low contrast areas [29], and the result will be obtained by applying this technique during the experimentation and evaluating its impact and further augmenting the diagnostic capabilities of the DINOv2 ViT model.

The subsequent sections will delve into the methodology, results, and implications of the research, offering a comprehensive understanding of how this innovative approach can revolutionize the classification of brain tumors.

2. Methodology.

2.1. Dataset description. This paper utilizes a composite dataset comprising three distinct sources: (1) “Brain MRI Images for Brain Tumor Detection” [30], (2) “Brain Tumor MRI Dataset” [31], and (3) “Br35H :: Brain Tumor Detection 2020” [32].

Within dataset (1), there are two classes denoted as “Yes” and “No.” “Yes” encompasses 155 images, while “No” comprises 98 images. In contrast, dataset (2) amalgamates three datasets: “Brain Tumor Dataset Version 5” [33], “Brain Tumor Classification (MRI)” [34], and “Br35H :: Brain Tumor Detection 2020” [32]. This dataset consists of four classes: glioma (1,621 images), meningioma (1,645 images), no-tumor (2,000 images), and pituitary (1,757 images). Dataset (3) also contains two classes, “Yes” (1,500 images) and “No” (1,500 images), similar to dataset (1).

Subsequently, these three datasets were amalgamated and grouped into two principal classes: “Tumor” and “No-Tumor”. The “Tumor” class incorporates datasets with “Yes”, glioma, meningioma, and pituitary classes, while the “No-Tumor” class is derived from datasets labeled “No”.

Consequently, the combined dataset encompasses a total of 10,276 brain MRI images; because 11 images were damaged, the total MRI images were 10,265 divided into two classes: 3,587 depicting No-Tumor, and 6,678 representing Tumor.

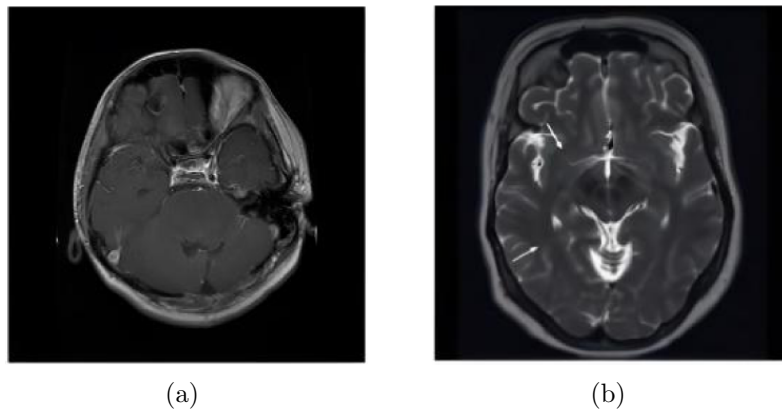


FIGURE 1. Sample MRI images for each class: (a) No-Tumor; (b) Tumor

The dataset is subsequently partitioned into training, validation, and testing subsets in a 70 : 10 : 20 ratio. The 7,182-image training set facilitates effective model learning, while the 1,026-image validation set supports hyperparameter tuning. The 2,057-image testing set ensures robust evaluation and statistical significance. The class-specific breakdown addresses imbalances, providing a strong foundation for medical imaging model development and evaluation, the details as shown in Table 1.

TABLE 1. Distribution of images for each class in the dataset

Class	Train	Val	Test
No-Tumor	2,508	359	720
Tumor	4,674	667	1,337
Total	7,182	1,026	2,057

This paper utilized techniques for image improvement, including histogram equalization, recognized as a highly effective approach to enhancing image quality [28]. Moreover, data augmentation techniques, including horizontal and vertical flipping, and rotation, were applied to the training dataset. These augmentations diversified the training data, mitigated overfitting, and improved the model’s generalization. This ensures consistency,

reduces complexity, preserves critical features, and enables the use of pre-trained models. Images are further standardized using mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225].

2.2. Experiment configuration. The classification task involved training six distinct model architectures over 50 epochs, employing the Adam optimizer with a learning rate of $1e-4$ [35], which has been proven suitable in other medical imaging studies, and setting a random seed of 42. Furthermore, early stopping was activated to mitigate overfitting. In the TL process, a fresh fully connected layer with randomized weights was introduced, and its output dimension matched the class count. Moreover, unfreezing specific pre-trained model layers for fine-tuning enhanced overall performance, impacting the count of trainable parameters, detailed in Table 2.

TABLE 2. Configuration of model parameters

Model	Parameters	Trainable parameters	Non-trainable parameters
DINOv2 ViT-S14	22,058,114	5,328,002	16,730,112
ViT-S32	22,880,490	5,710,698	17,169,792
SimCLRv2-ResNet50	37,266,122	4,283,016	32,983,106
ResNet50	25,558,570	6,513,130	19,045,440
VGG-19	143,668,778	4,098,538	139,570,240
EfficientNetB0	5,290,086	4,007,548	1,282,538

The parameter distribution signifies effective TL, emphasizing adaptability with a new fully connected layer and unfreezing pre-trained layers. These insights guide considerations, spotlighting trade-offs in complexity, resource efficiency, and effective TL for the task.

2.3. Model evaluation. To evaluate the model’s performance, key metrics such as precision (P), recall (R), and F1-score (F) were employed. These metrics are calculated using True Positives (TP), False Positives (FP), and False Negatives (FN), as indicated in Equation (1).

$$F = \frac{2P \cdot R}{P + R}, \text{ where } P = \frac{TP}{TP + FP}, R = \frac{TP}{TP + FN} \quad (1)$$

ROC AUC (Receiver Operating Characteristic – Area under the Curve) stands as a valuable metric for assessing the performance of the proposed approach. It gauges the model’s ability to differentiate between positive and negative classes. Understanding the model’s prediction rationale is crucial for reliable results.

The evaluation techniques were applied to the proposed model, DINOv2 ViT, along with supervised ViT, SimCLRv2-ResNet50, and ResNet50, evaluated and compared to baselines ResNet50, EfficientNetB0, and VGG-19. The baselines, pre-trained on ImageNet, unfroze the last three blocks for ResNet50 and VGG-19, and only the last layer for EfficientNetB0, with consistent hyperparameter settings.

3. Result and Discussion. It can be seen in Table 3, which shows that the DINOv2 ViT-S14 is the best model, as shown by the F1-score at 99.84%. With the problem of an imbalanced dataset, the models generally produce good performance.

All models leveraging SimCLRv2 or DINOv2 consistently outperform traditional supervised models, demonstrating the effectiveness of these methods in addressing issues associated with dataset imbalance. The high AUC scores across all models signify robust discriminative ability as shown in Table 3.

TABLE 3. Evaluation result on the test set

Model	Pre (%)	Rec (%)	F1 (%)	AUC
DINOv2 ViT-S14	99.79	99.89	99.84	1.00
ViT-S32	95.20	95.43	95.31	0.99
SimCLRv2-ResNet50	99.74	99.62	99.68	1.00
ResNet50	97.30	97.36	97.33	1.00
VGG-19	89.81	90.74	90.24	0.97
EfficientNetB0	93.45	94.51	93.93	0.99

Figure 2 shows the confusion matrix and its extent under the Receiver Operating Characteristic (ROC) curve for DINOv2 ViT-S14 for imbalanced test data. Figure 2(a) shows that the model can predict No-Tumor and Tumor with less than 1% error, 720 of 720 images for No-Tumor, and 1,334 of 1,337 images for Tumor. This shows that the model is robust enough to handle imbalanced data, considering the limited availability of data. The model's overall performance looks promising, as evidenced by its F1-score. Figure 2(b) shows the area under the ROC curve for one vs rest approach for all models. The DINOv2 ViT-S14 achieved a perfect score of 1.00 for classifying No-Tumor and Tumor, which indicates excellent discrimination by the model. It excels at distinguishing these two classes with high accuracy and without any false positives or false negatives. SimCLRv2-ResNet50 and ResNet50 closely follow with an AUC of 1.00. These models show strong discrimination, with DINOv2 ViT-S14 identified as the best using F1-score and AUC.

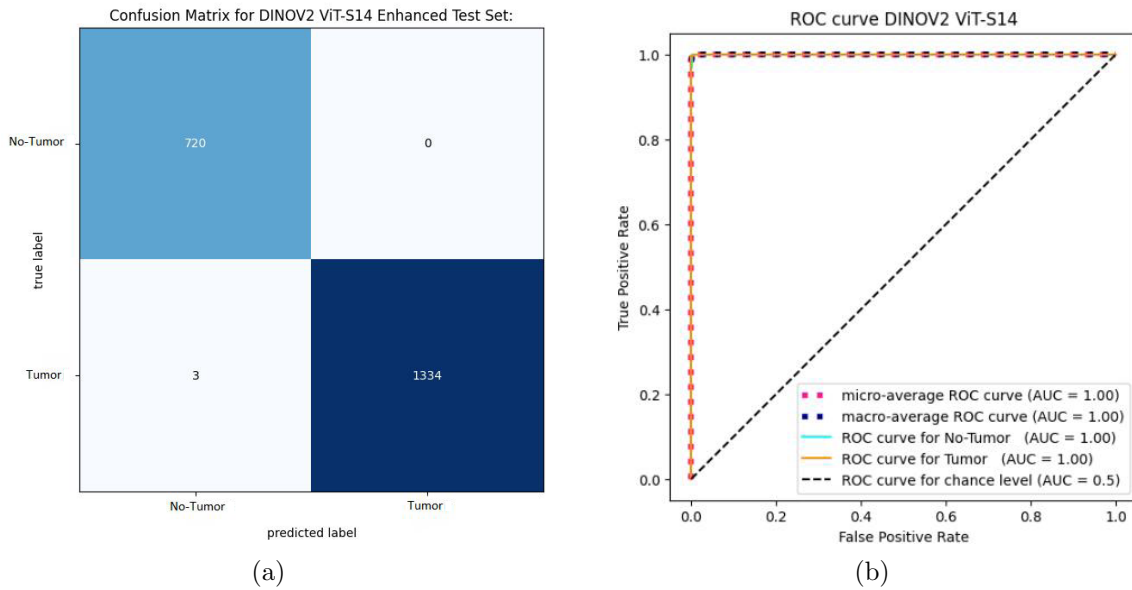


FIGURE 2. Confusion matrix (a) and ROC AUC curve (b) for each class

4. Conclusion. This paper proposes a ViT model pre-trained with DINO to classify brain tumors from MRI images. The performance of the model is compared with the ResNet50 model, which has been pre-trained with SimCLRv2, with other supervised models such as ResNet50, VGG-19, and EfficientNetB0. The DINOv2 ViT-S14 model shows excellent results, even when dealing with imbalanced data. The proposed model surpasses prior approaches, showing potential for high accuracy, particularly suitable for medical applications. The ROC AUC curve demonstrates SSL's superior performance over the supervised model, highlighting its effectiveness with imbalanced data and can be applied to various medical image classifications.

A limitation of this study is the imbalance of the dataset; even though it exhibits a flawless ROC curve, this perfection may be overly optimistic and could prove to be an unreliable benchmark when compared to present-day data. Furthermore, this research focuses on using MRI images to achieve the highest level of brain tumor classification, among other types of images but only in binary class. To address this limitation, future research should engage in multiclass classification for classifying different brain tumor types. Additionally, consider enlarging the training dataset through techniques such as data augmentation and oversampling while exploring alternative methods for evaluating model predictions.

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